

Rapport Modélisation Prédictive

Anne-Laure Sulmont et Amjad Proietti

Contents

1	Introduction	2
2	Data Analysis and Enrichment	2
2.1	Temporal Dynamics and Thermosensitivity	2
2.2	Open Data Enrichment: The Price Shock (ENTSO-E)	3
3	Seperating our models	3
3.1	Predicting Quantiles	4
3.2	Comparing models	4
4	Standard model	5
5	Residual Correction	6
6	QGAM	9
7	Expert Aggregation	9
8	Conclusion	10
9	Bibliography	12

1 Introduction

The objective of this predictive modeling project is to estimate as accurately as possible the Net Demand for electricity in France during the period known as *energy sobriety*. Net Demand represents the total electricity consumption minus the production from non-dispatchable renewable energy sources (wind and solar). It therefore corresponds to the share of demand that must be supplied by dispatchable generation sources such as nuclear, hydropower, gas, and coal.

Predicting Net Demand is crucial for several reasons. In the current energy context—characterized by strong price volatility and supply tensions (particularly after the 2022 energy crisis), anticipating electricity needs is a matter of public security, as it helps prevent power outages (blackouts). It is also an economic and environmental imperative: poor forecasts may force operators to activate thermal power plants on short notice, which are both very costly and highly CO₂-emitting, in order to balance the electricity grid.

To evaluate the performance of our predictions, the metric imposed for this project is the minimization of the Pinball Loss function at the 80th quantile ($\tau = 0.8$) over the test period.

Unlike a classical quadratic error (RMSE), which targets the mean and penalizes errors symmetrically, quantile regression aims to estimate a conditional quantile:

$$\mu_\tau(x) = F^{-1}(\tau \mid x)$$

The associated loss function is defined as:

$$\rho_\tau(z) = \begin{cases} \tau z & \text{if } z \geq 0, \\ (\tau - 1)z & \text{if } z < 0. \end{cases}$$

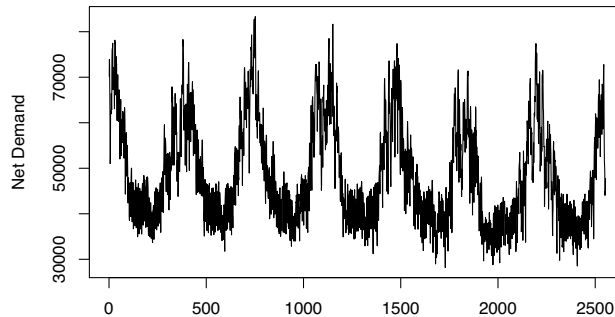
Where z represents the prediction error ($y_{\text{true}} - y_{\text{pred}}$). The choice of the quantile $\tau = 0.80$ is not arbitrary: it means that our model will penalize underestimation of demand ($z \geq 0$) four times more than overestimation ($z < 0$). This reflects the reality of electricity system operations, where the risk of energy shortage is considered far more critical than a slight surplus of production.

2 Data Analysis and Enrichment

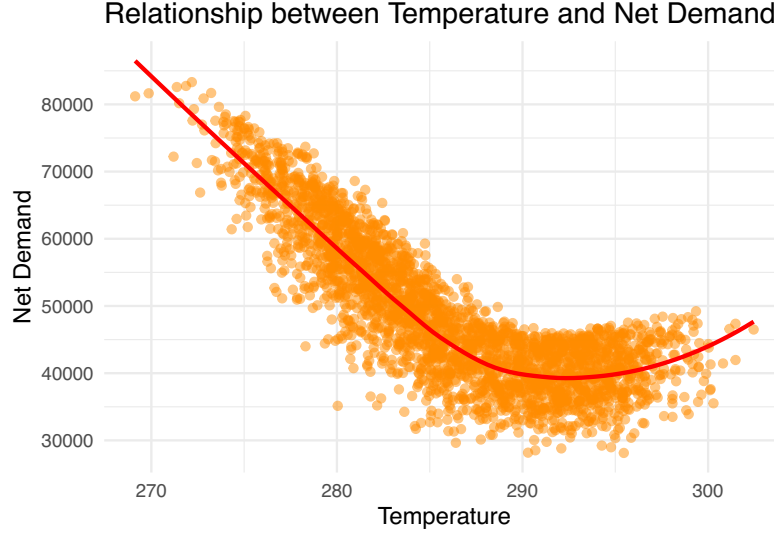
2.1 Temporal Dynamics and Thermosensitivity

The training dataset spans from March 2013 to September 2022 and contains the target variable Net_demand along with numerous explanatory variables that we structured into six major thematic categories: calendar variables capturing human activity (public holidays, weekends, vacations); temperature variables, the predominant driver of electricity consumption, seasonal dynamic variables (*toy*, *DLS*); lags from the previous day and the previous week; solar impact variables and wind energy variables.

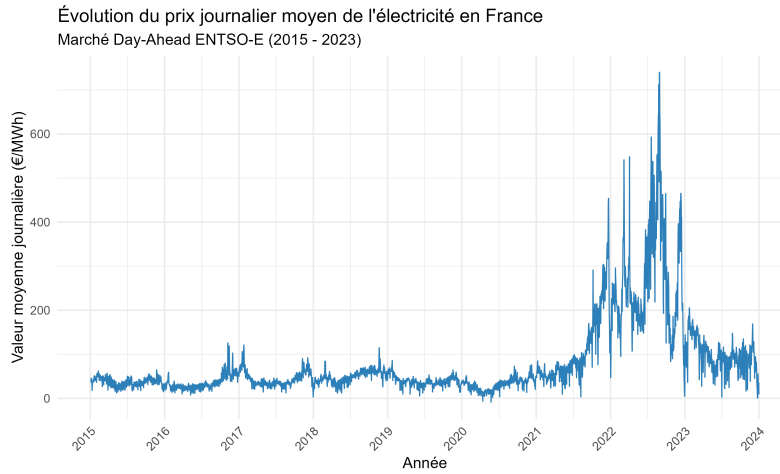
A visual analysis of the target variable reveals strong annual, seasonal and weekly cycles.



Furthermore, examining the relationship between temperature and electricity demand highlights a strong non-linear relationship. This is why we quickly abandoned our linear regression model.



2.2 Open Data Enrichment: The Price Shock (ENTSO-E)



2022 was marked by an energy crisis, showing an unprecedented surge and volatility in daily prices on the Day-Ahead electricity market. Based on the hypothesis that these price shocks may alter consumer behavior, we integrated ENTSO-E price data into our variables.^[1]

We computed general statistics (mean price, maximum, minimum, and standard deviation) to capture the overall market tension. In addition, we created period-specific indicators such as evening average prices, evening peaks, and nighttime prices, in order to isolate periods of high electricity consumption.

Finally, we included dynamic and volatility variables, such as price ramps (ramp_18_6, ramp_max), in order to quantify the intensity of abrupt price changes from one hour to the next.

3 Seperating our models

We took inspiration from our classmates and decided to split our models of Net demand using the following formula.

$$NetDemand = Load - WindPower - SolarPower$$

By modeling separately for load, solar power and wind power, we believed that it would lead to stronger predictions, since we could tailor the variables chosen to each model, making them better. After the analysis of our data, clearly the relationship between the quantitative variables and each of the targets were often non-linear. We thus decided to use Generalized Additive Models (GAMs), since they allow to predict for linear and non-linear relationships. The model can be written as:

$$g(\mathbb{E}[Y | X]) = \beta_0 + \sum_{j=1}^p \beta_j X_j + \sum_{k=1}^K f_k(X_k)$$

where: - Y is the response variable, - $g(\cdot)$ is the link function, - β_0 is the intercept, - $\beta_j X_j$ represents the linear effects of some predictors, - $f_k(X_k)$ denotes the smooth non-linear effects estimated from the data.

3.1 Predicting Quantiles

Since GAMs estimate conditional means rather than conditional quantiles, we transformed the mean predictions into quantile forecasts using two approaches. First, we assumed normally distributed residuals and computed the quantile forecast as $q_\alpha = \mu + z_\alpha \sigma$. Second, to avoid relying on a normality assumption, we used the empirical quantile of the training residuals. In both cases, the quantile correction was added to the mean prediction.

3.2 Comparing models

To compare model performance, we used rolling-block cross-validation in order to preserve the temporal dependence structure of the data. Each training block covered one year, while each test block covered six months, with six-month increments between successive splits. For each block, we fitted the GAM, predicted on the test set, applied the quantile correction, and computed the pinball loss. The function can be found at the end of our code for our final model.

In order to identify relevant predictors, we combined several approaches. First, we fitted a lasso model on all available variables and used $\lambda = \lambda_{1se}$ to identify variables shrunk to zero. Although lasso optimizes mean squared error rather than pinball loss, it provided a useful starting point for selection.

We then complemented this with our knowledge and the double-penalty approach available in `mgcv` through `select = TRUE`, which penalizes both linear and smooth terms. The model can be written as:

$$g(\mathbb{E}[Y]) = \beta_0 + \sum_{j=1}^p \beta_j X_j + \sum_{k=1}^K f_k(X_k)$$

where each smooth function $f_k(\cdot)$ is estimated using a double-penalty structure:

$$\lambda_{k1} \int (f_k''(x))^2 dx + \lambda_{k2} \|f_k\|^2$$

where:

- λ_{k1} is the wiggleness penalty controlling smoothness,
- λ_{k2} is the null-space penalty, allowing the entire smooth term to shrink toward zero.

Finally, when choosing our hyper-parameters `bs`, we took “cr” for most of our variables, as suggested by our professor. The only exception was for the time of year variables, which goes from 0 to 1, modelling where we are in the year. We used “cc” since this allows the spline to end where it starting well simulating the cycle of the variable.

3.2.1 Load

Unexpectedly, the load model did not improve when electricity price variables were included, whereas solar and wind models did.

Our final formula for load was:

$$Load = \beta_0 + s_{cc}(toy) + s_{cr}(Temp) + s_{cr}(Load_{t-1}) + \beta_1 WeekDays + \varepsilon$$

```
summary(gam_load)$s.table
```

##		edf	Ref.df	F	p-value
##	s(toy)	26.245727	28.000000	15.61219	0
##	s(Temp)	5.479018	6.664665	362.53629	0
##	s(Load.1)	8.770417	8.974948	621.95694	0

From the effective degrees of freedom (edf), we found that the time-of-year variable required a larger basis dimension. In practice, $k \geq 30$ was needed for `toy`, while $k = 10$ was sufficient for the other smooth terms. The same conclusion also applied to the solar and wind models.

3.2.2 Solar and Wind Power

For both solar and wind power, nebulosity quickly appeared to be an important predictor. However, because the data collection methodology changed in 2017, we compared several training strategies: using the full dataset, using the full dataset with an interaction between nebulosity and time of year, and using only post-2017 observations.

The best results for both models were obtained by training only on data from 2017 onward, with an interaction in solar power. Interestingly, the variables selected by lasso also gave the strongest performance in the GAM framework.

$$\begin{aligned} Solar_power = & s(toy) + s(Temp) + s(Temp_s95_min) + s(Temp_s99_min) + s(Temp_s99_max) \\ & + s(Wind) + s(Wind_weighted) + s(Solar_power_{t-1}) + te(Nebulosity, toy) + s(Solar_power_{t-7}) \\ & + \beta_1 Weekdays + \beta_2 BH_before + \beta_3 BH + \beta_4 Year + \beta_5 Month + \beta_6 DLS + \beta_7 Summer_break \\ & + \beta_8 Christmas_break + \beta_9 Holiday + \beta_{10} Holiday_zone_c + s(Wind_power_{t-1}) \\ & + s(Wind_power_{t-7}) + s(Net_demand_{t-1}) + s(mean) + s(night_mean) + s(std) + \varepsilon \end{aligned}$$

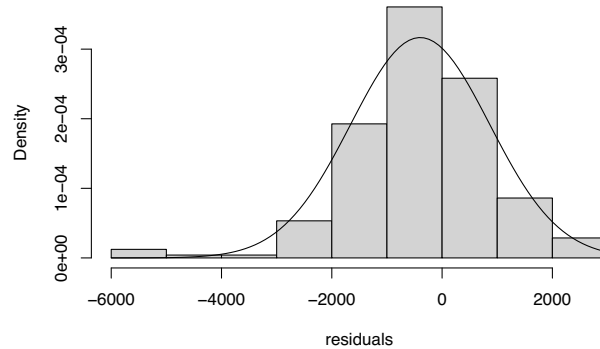
$$\begin{aligned} Wind_power = & s(toy) + s(Temp) + s(Wind_power_{t-1}) + s(Wind_power_{t-7}) + s(Wind) \\ & + s(Load_{t-7}) + s(Temp_s99_min) + s(Wind_weighted) + s(Nebulosity) \\ & + \beta_1 WeekDays + \beta_2 DLS + \beta_3 Summer_break + \beta_4 Holiday_zone_a + \beta_5 Holiday_zone_b \\ & + \beta_6 BH_Holiday + s(Solar_power_{t-1}) + s(Solar_power_{t-7}) + s(min) + s(std) + \varepsilon \end{aligned}$$

4 Standard model

Once we obtained the models that produced the best mean predictions for load, solar power, and wind power, we combined these predictions before performing any residual correction. Our idea was that the prediction errors of the three components could be dependent, and therefore it was more coherent to treat them jointly at the level of net demand.

```
# Net demand mean prediction and residuals on validation set
forecast.3mod.validation = pred_load_validation - pred_wind_validation - pred_solar_validation
residuals = validation$Net_demand - forecast.3mod.validation
```

We used the same two methods to obtain quantile predictions. We checked that our final residuals follow approximately a normal distribution using the histogram.

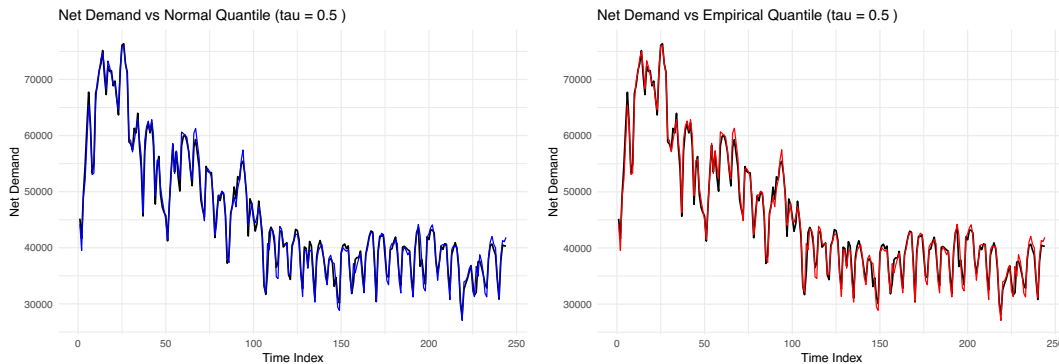


```
sigma_hat = sd(residuals, na.rm=T)
mu_bias = mean(residuals, na.rm=T)

quantile.forecast.3mod.validation = forecast.3mod.validation + mu_bias +
  qnorm(tau) * sigma_hat
```

We also computed empirical quantile predictions using the empirical distribution of the residuals, without assuming normality.

```
emp.quant = quantile(residuals, probs = tau, na.rm=T)
emp.quant.fore.3mod.validation = forecast.3mod.validation + emp.quant
```

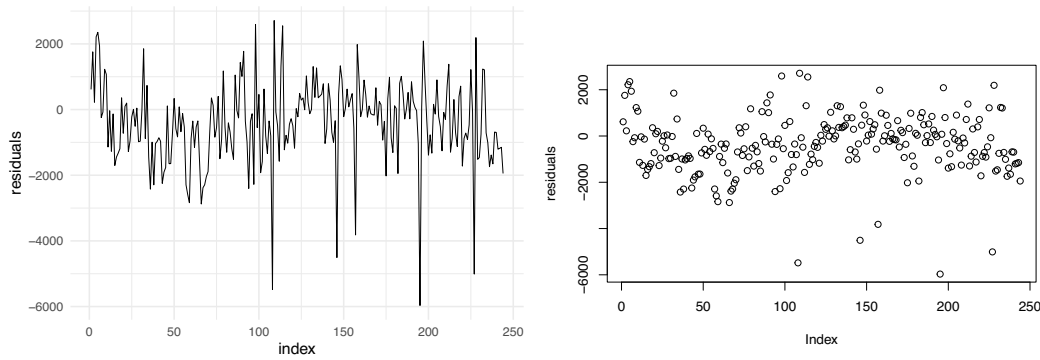


Both approaches of our standard model produced very similar results, suggesting that the normality assumption for the residuals was reasonable. It even yielded the best performance, achieving a public score of 416 on Kaggle.

To further improve this score, we examined the residuals to determine whether applying an ARIMA model would be an appropriate next step.

5 Residual Correction

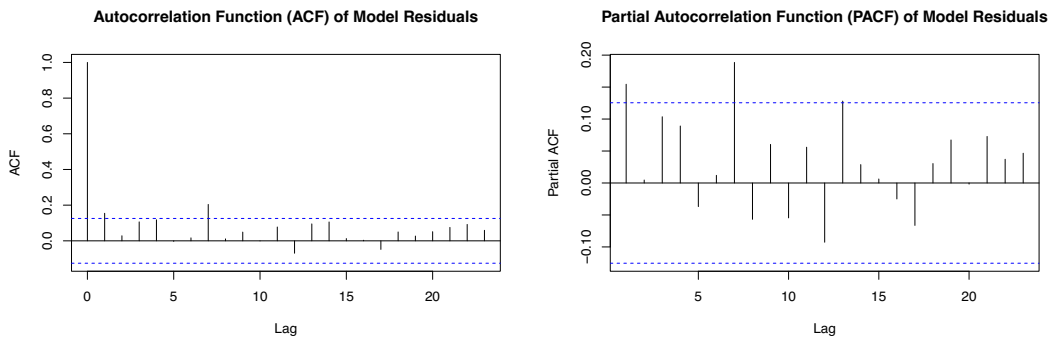
By plotting our residuals, we observed that they are not white noise and still exhibit temporal dependence. In addition, the mean of the residuals is not constant, dropping to around 50 at certain periods and toward the end of the series. The variance also appears to change over time, suggesting that the process may not be stationary. This motivated us to explore whether an ARIMA (Auto-Regressive Integrated Moving Average) model could better capture the remaining structure in the residuals.



To formally test for autocorrelation, we performed the Box–Ljung test. This test evaluates the null hypothesis that the residuals have no autocorrelation against the alternative. Our p-value was below 0.05, leading us to reject the null hypothesis and conclude that there is still unexplained dependence in the residuals.

```
##
## Box-Ljung test
##
## data: residuals
## X-squared = 33.92, df = 20, p-value = 0.02667
```

We then examined the autocorrelation function (ACF), which measures the correlation between the series and its lagged values at different time lags. The ACF revealed that at lag 7 the correlation exceeds the 95% confidence bounds (represented by the blue lines), indicating statistically significant autocorrelation at this lag. Furthermore, the gradual geometric decay of the autocorrelation after lag 7 suggests that an autoregressive component may be appropriate.

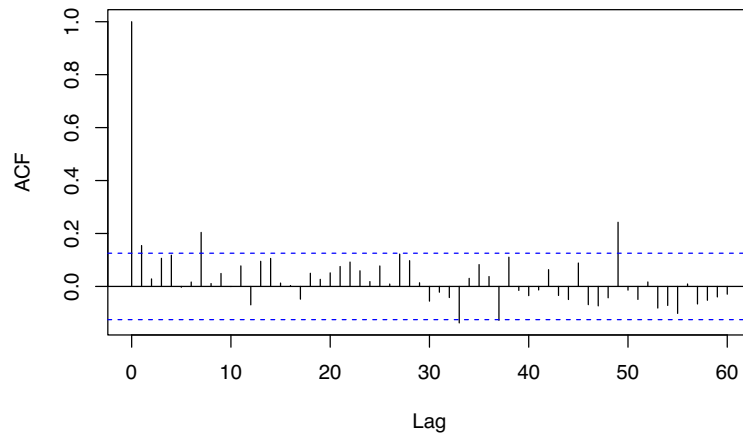


We also inspected the partial autocorrelation function (PACF), which measures the direct correlation between the series and a given lag after removing the effect of intermediate lags. Similarly, we observe a significant spike at lag 7, suggesting that a weekly influence remains in the residuals and was not fully captured by our model.

However, after this lag, the PACF does not display a sharp cut-off but rather a gradual decay. This pattern suggests that, in addition to an autoregressive component, a Moving Average (MA) component may also be useful for modeling the remaining dependence structure.

Finally, we examined the ACF up to a lag of 60 days to determine whether a seasonal ARIMA (SARIMA) model might be appropriate. However, we did not observe any clear seasonal spikes or repeating lag patterns in the ACF, which led us to conclude that a seasonal component was unnecessary in our case.

Autocorrelation Function of Residuals (60 Lags)

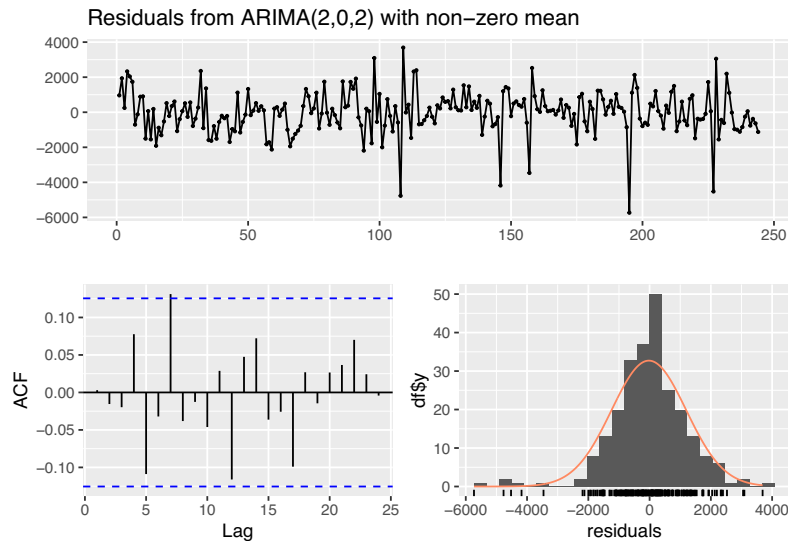


Overall, these results suggest that modeling the residuals with an ARIMA process could potentially improve the predictions. However, the diagnostic indicators do not show a very strong autocorrelation structure, which suggests that the benefits of introducing an ARIMA component may be limited. We used the `auto.arima()` function from the `forecast` package to perform a grid search in order to identify the best ARIMA coefficients.

We selected the ARIMA model using the AIC criterion since unlike the BIC, which penalizes model complexity more strongly, the AIC allows for slightly more flexible models. We favored this since we want to have the best generalization performance.

In our summary, we find that the best model is an ARMA model with AR(2) and MA(2) components and no differentiating. This indicates that the original series was already sufficiently stationary.

```
checkresiduals(fit_auto)
```



```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(2,0,2) with non-zero mean
## Q* = 10.218, df = 6, p-value = 0.1158
##
## Model df: 4.   Total lags used: 10
```

We used a residual diagnostic to evaluate the adequacy of the model. From the residual plot, the mean of the residuals appears approximately constant and centered around zero, without the large shifts seen previously. The variance still shows some fluctuation, but no clear systematic pattern.

Examining the ACF of the residuals, only one lag slightly exceeds the confidence interval, suggesting that most of the autocorrelation has been removed by the model. This is further supported by the Ljung–Box test, which produces a p-value greater than 0.5. We thus fail to reject the null hypothesis that the residuals are uncorrelated, indicating that the ARMA(2,2) model adequately captures the dependence structure of the data.

We then used our best ARMA coefficients to forecast on our validation set.

When comparing the pinball loss on the validation set before and after applying the ARIMA correction, we observed that the pinball loss got worse. On Kaggle, the public leaderboard score was 417 for both the normal quantile and empirical quantile approaches, which is not better than our previous results.

##	Model	Quantile_Type	Pinball_Loss
## 1	Standard Model	Normal Quantiles	464.9651
## 2	Standard Model	Empirical Quantiles	463.8682
## 3	ARMA Model	Normal Quantiles	467.5996
## 4	ARMA Model	Empirical Quantiles	467.0639

Although it was reasonable to attempt an ARMA correction to remove the remaining autocorrelation in the residuals, and although this procedure did reduce the residual dependence, it ultimately did not improve the forecasting performance of our model. Therefore, incorporating an ARIMA model did not provide a meaningful improvement for the final predictions.

6 QGAM

In order to compare our previous models with a more standard approach, we fitted a Quantile Generalized Additive Model (QGAM) directly on Net Demand. This model follows the same principle as a GAM, but instead of predicting the conditional mean, it directly estimates the desired quantiles of the response variable. As before, we experimented with several approaches for variable selection. However, the variables selected using the lasso procedure again produced the best results.

Among the different quantiles tested, the median prediction ($\tau = 0.5$) performed best on Kaggle, yielding a public leaderboard score of 460. While this result was still reasonable, it confirmed that our strategy of modeling load, wind power, and solar power separately led to better overall performance.

7 Expert Aggregation

After building the previous models, we were introduced in class to the `opera` library. This package allows the construction of ensemble models by combining several forecasting models into a single aggregated predictor. The method relies on online learning, meaning that the weights assigned to each model are updated sequentially over time according to their predictive performance. We chose to use have the pinball loss be minimized by the experts, since that is the performance evaluation we have been working with.

We included the following models in the ensemble: our standard GAM-based model with both normal and empirical quantile corrections, the ARMA-corrected model with normal and empirical quantiles, and the QGAM model.

Up to this point, we had used $\tau = 0.5$ for all quantile predictions, as this value provided the best performance on Kaggle. However, to take full advantage of the expert aggregation framework, we investigated whether predicting different quantiles with each model would affect the ensemble’s performance. In particular, we evaluated how the choice of quantile influenced both the cumulative loss and the weights assigned to each model.

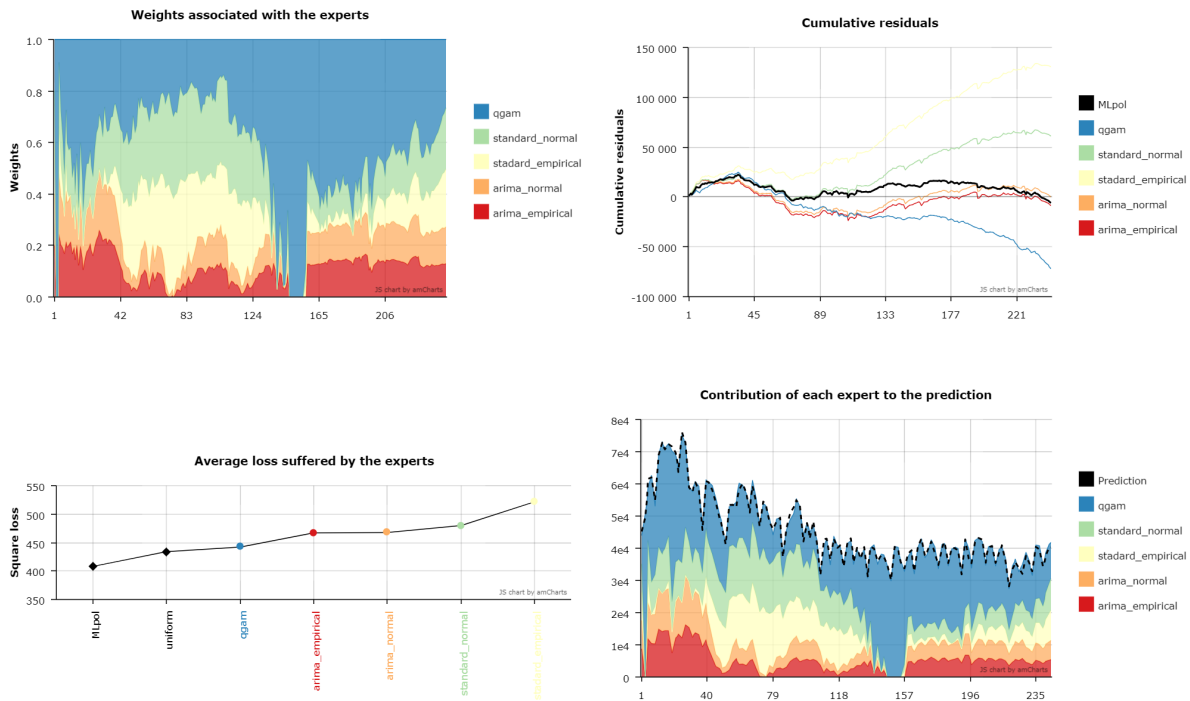
We therefore tested three different quantiles: $\tau = 0.3$, $\tau = 0.5$, and $\tau = 0.8$. For each value of τ , we ran the same modeling pipeline described above, simply modifying the quantile level used in the prediction step.

Our first finding was that the best ensemble performance occurred when all models predicted the 30% quantile ($\tau = 0.3$). We then extended the experiment by allowing different models to predict different quantiles. The

best-performing configuration was obtained when the standard model and the QGAM predicted the 30% quantile, while the ARMA model predicted the median quantile ($\tau = 0.5$).

```
experts <- cbind(forecast.qgam.validation.3,
                quantile.forecast.3mod.validation.3,
                quantile.forecast.with.arima.validation.5,
                emp.quant.fore.3mod.validation.3,
                emp.quant.fore.arima.validation.5)
colnames(experts) <- c("qgam", "standard_normal", "arima_normal",
                      "standard_empirical", "arima_empirical")

agg <- mixture(Y=validation$Net_demand, experts=experts,
              loss.type = list(name='pinball', tau=0.5), model = "MLpol")
```



By examining the plots, we observe that the ensemble model achieves the lowest pinball loss and cumulative residuals compared to the individual models. This indicates that the aggregation approach improves predictive performance, thus supporting our decision to combine the models using expert aggregation.

Additionally, the evolution of the weights shows that the different models maintain relatively similar importance throughout the learning process. This suggests that each model contributes useful information to the final prediction, and that no single model consistently dominates the ensemble.

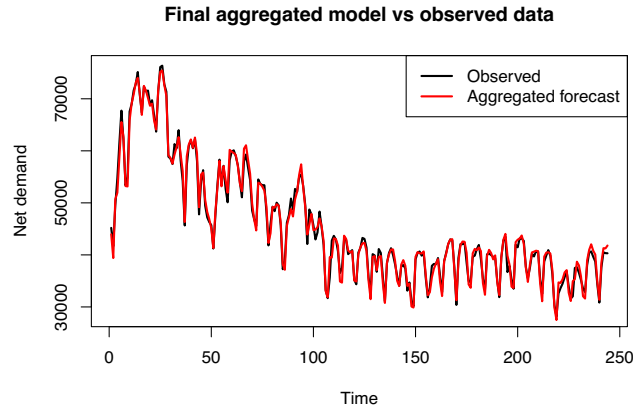
8 Conclusion

Final Model Performance:

Here we plot our final predictions on the validation set against the observed Net Demand values. This visualization allows us to assess how closely the predictions follow the true series and whether the model successfully captures the underlying dynamics of the data.

Overall, the predictions track the observed values relatively well. In particular, the model appears to capture both the seasonal patterns and the general trend present in the time series.

On Kaggle, our final model achieved a private score of 402 and a public score of 414. The difference between these scores suggests that the model slightly overfitted the validation data used during development. Nevertheless, the final approach still represents a small improvement over our baseline model.



Limitations and Possible Improvements:

Although we are satisfied with the final performance of our approach, several aspects of the model could be improved further.

First, we may have combined the individual models too early in the modeling process. Instead of aggregating them at an early stage, it could have been beneficial to perform residual correction on each model separately before combining them.

Second, the residual diagnostics after fitting the ARMA model suggest that some temporal dependence may still remain. In particular, one lag slightly exceeds the confidence interval, indicating that the residuals are not entirely decorrelated. This suggests that an additional ARIMA-type correction on the new residuals could have been explored to further reduce any remaining autocorrelation.

Finally, the models included in the expert aggregation are relatively similar in nature, which may limit the diversity of the ensemble. Since ensemble methods often benefit from combining models that capture different patterns in the data, performance might have improved by incorporating more heterogeneous learners, such as gradient boosting methods.

9 Bibliography

1. European Network of Transmission System Operators for Electricity (ENTSO-E). *Day-Ahead Energy Prices – France*. Available at: <https://tp.entsoe.eu/usy-tp-guig02/7ee182f033dd5b139967bb1434ddaa84/market/energyPrices> (Accessed: 10 February 2026).